

User Acceptance Test — Core Flows

InvoiceDirect Mass Mailer System

PEPPOL Compliant Invoice Delivery system)

Version: 1.0

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Focus: End-to-End Invoice Dispatch Flow

1. Overview

This document covers the three core business flows for UAT:

- Flow A — Fiscalised Invoice → PDF Watcher → Email Delivery (end-to-end dispatch with ZIMRA FDMS validation, status tracking, and delivery monitoring)
- Flow B — Customer Creation via UI
- Flow C — Customer Import via CSV

2. Status Key

Status	Meaning
Pass	Actual result matches expected result
Fail	Actual result does not match expected result
Blocked	Could not execute due to dependency or environment issue

Flow A: Fiscalised Invoice → PDF Watcher → Email Delivery

A-01: PDF Watcher picks up fiscalised PDF with ZIMRA FDMS markers

Priority	Critical	Feature Ref	Flow A
Description	Verify: pdf watcher picks up fiscalised pdf with zimra fdms markers	Status	Pass / Fail / Blocked

Test Steps:

1. Preconditions
 - PDF_WATCHER_ENABLED=true
 - inbox directory exists and is writable
 - SMTP configured, test mailbox ready
2. Drop a valid ZIMRA-fiscalised PDF (containing fdms.zimra.co.zw, verification code, device ID) + matching sidecar JSON into inbox
3. Wait one poll interval (~5-10s)

Expected Result:

- Watcher detects ENTRY_CREATE event for .pdf file.
- Watcher log shows "Processing fiscalised invoice PDF: {filename}".
- ZimraFiscalValidator validates: (a) fdms.zimra.co.zw domain present, (b) verification code found, (c) fiscal device field present.
- Validation passes: "All ZIMRA fiscal markers present".
- PdfAttachmentResolver confirms valid PDF (%PDF- magic bytes).
- Sidecar JSON is parsed for invoice metadata.

A-02: Customer auto-upserted from Buyer block before dispatch

Priority	Critical	Feature Ref	Flow A
Description	Verify: customer auto-upserted from buyer block before dispatch	Status	Pass / Fail / Blocked

Test Steps:

1. Confirm the buyer email from the PDF is NOT already in the customers table
2. Drop the fiscalised PDF + sidecar into inbox
3. After processing, query the customer list

Expected Result:

- A new Customer record is auto-created for the buyer (email extracted from PDF Buyer block or sidecar).
- erpSource is set to indicate the source (e.g., PDF_WATCHER).
- If sidecar contains VAT/TIN, PEPPOL participant ID is derived as 0190:ZW{vat}.
- The customer appears immediately in the Customers page.
- If buyer already existed, existing record is updated (not duplicated).

A-03: PDF attached to email and sent to recipient

Priority	Critical	Feature Ref	Flow A
Description	Verify: pdf attached to email and sent to recipient	Status	Pass / Fail / Blocked

Test Steps:

1. After watcher processes the PDF (A-01 passed)
2. Check the recipient's email inbox
3. Verify email headers and attachment

Expected Result:

- Email is received in the recipient inbox.
- From address matches the configured sender.
- Subject contains the invoice number.
- Body contains invoice metadata (company, amount, fiscal fields) rendered via template.
- PDF attachment is present with filename = {invoiceNumber}.pdf.
- Attachment opens correctly and displays the original fiscalised invoice.
- If customer has multiple contacts, sibling contacts receive CC with same PDF.

Notes: Verify both SMTP API (if enabled) and SMTP fallback paths.

A-04: MailRecipient status set to SENT/DELIVERED after successful dispatch

Priority	Critical	Feature Ref	Flow A
Description	Verify: mailrecipient status set to sent/delivered after successful dispatch	Status	Pass / Fail / Blocked

Test Steps:

10. 1. After A-03 confirms email sent
11. 2. Poll GET /api/v1/campaigns/{campaignId}
12. 3. Query database: SELECT status, message_id, sent_at, attachment_size_bytes FROM mail_recipient WHERE invoice_number = ...

Expected Result:

- MailRecipient.status = "SENT" or "DELIVERED".
- messageId is populated (non-null SMTP/Brevo message ID).
- sentAt timestamp is populated and is recent.
- attachmentSizeBytes matches the original PDF size.
- Campaign.sentCount is incremented.
- MeteringService recorded a UsageRecord for this delivery.

A-05: Campaign status shows COMPLETED after all recipients processed

Priority	High	Feature Ref	Flow A
Description	Verify: campaign status shows completed after all recipients processed	Status	Pass / Fail / Blocked

Test Steps:

13. 1. Drop 3 fiscalised PDFs into inbox (same batch)
14. 2. Poll GET /api/v1/campaigns/{campaignId} until processing finishes
15. 3. Observe status transitions

Expected Result:

- Campaign status transitions: CREATED → QUEUED → IN_PROGRESS → COMPLETED.
- sentCount = 3, failedCount = 0, skippedCount = 0.
- completedAt timestamp is populated.
- startedAt is before completedAt.
- Total recipients count matches the number of PDFs dropped.

A-06: PDF missing ZIMRA FDMS markers is rejected by watcher

Priority	Critical	Feature Ref	Flow A
Description	Verify: pdf missing zimra fdms markers is rejected by watcher	Status	Pass / Fail / Blocked

Test Steps:

16. 1. Generate a PDF that is a valid PDF (%PDF- header) but contains NO ZIMRA FDMS markers
 - No "fdms.zimra.co.zw" string
 - No verification code (no 16-char hex pattern, no "Verification Code" label)
 - No fiscal device fields
17. 2. Drop it into inbox with sidecar JSON
18. 3. Wait for processing

Expected Result:

- Watcher picks up the PDF.
- ZimraFiscalValidator validates and FAILS.
- Validation errors logged: "Missing FDMS domain", "Missing verification code", "Missing fiscal device field".
- PDF is NOT emailed to any recipient.
- PDF + sidecar are moved to inbox/failed/ directory.
- No campaign or MailRecipient records are created.
- Error is logged with specific details of missing markers.

Notes: Use a PDF library to strip fiscal markers from a known-good ZIMRA PDF for this test.

A-07: MailRecipient status set to FAILED on email delivery error

Priority	High	Feature Ref	Flow A
Description	Verify: mailrecipient status set to failed on email delivery error	Status	Pass / Fail / Blocked

Test Steps:

19. 1. Configure SMTP to point to a non-existent/inaccessible server (or use an invalid recipient domain)
20. 2. Drop a valid fiscalised PDF + sidecar into inbox
21. 3. Wait for processing
22. 4. Poll campaign status
23. 5. Query MailRecipient

Expected Result:

- MailRecipient.status = "FAILED".
- errorMessage describes the failure reason (e.g., "Mail server connection failed" or "Invalid recipient domain").
- retryCount = 0 (first attempt).
- Campaign status = "PARTIALLY_FAILED" (if others succeeded) or "FAILED".
- MeteringService records a UsageRecord with outcome=failed.

Notes: Restore correct SMTP config after this test. @Retryable will attempt 3 retries with exponential backoff before final FAILED status.

A-08: Retry failed recipients — status resets to PENDING and re-dispatches

Priority	High	Feature Ref	Flow A
Description	Verify: retry failed recipients — status resets to pending and re-dispatches	Status	Pass / Fail / Blocked

Test Steps:

24. 1. Cause a delivery failure (see A-07)
25. 2. POST /api/v1/campaigns/{campaignId}/retry
26. 3. Poll campaign status until processing completes
27. 4. Query MailRecipient

Expected Result:

- On retry: failed MailRecipient rows are reset to status = "PENDING".
- Dispatch pipeline re-runs for those recipients.
- If the retry succeeds: status = "SENT", retryCount incremented.
- If the retry fails again: status stays "FAILED", retryCount incremented.
- After maxRetries (default 3) exceeded, subsequent retries are skipped.
- The POST endpoint returns immediately (async); polling tracks progress.

A-09: PDF moved to processed/emailed archive after successful dispatch

Priority	Medium	Feature Ref	Flow A
Description	Verify: pdf moved to processed/emailed archive after successful dispatch	Status	Pass / Fail / Blocked

Test Steps:

28. 1. Drop a valid fiscalised PDF + sidecar into inbox
29. 2. Wait for successful processing (A-01 through A-04 pass)
30. 3. Check the inbox directory structure

Expected Result:

- Original PDF is moved from inbox/ to inbox/processed/.
- Sidecar JSON is also moved to inbox/processed/.
- A copy is placed in the emailedDirectory archive (if configured).
- Timestamp is preserved in the filename or file metadata.
- On failure: files moved to inbox/failed/ instead.

Flow B: Customer Creation via UI

B-01: Create customer with single contact via UI Register Customer form

Priority	Critical	Feature Ref	Flow B
Description	Verify: create customer with single contact via ui register customer form	Status	Pass / Fail / Blocked

Test Steps:

31. 1. Log in to admin interface
32. 2. Navigate to Customers page
33. 3. Click "Register Customer" button
34. 4. Fill in: Company Name, erpCustomerId, Contact Email, Contact Name
35. 5. Optionally set: delivery mode (EMAIL/AS4/BOTH), VAT/TIN
36. 6. Save

Expected Result:

- Customer appears at the top of the customer list.
- Detail panel shows the customer with company name, erpCustomerId, and the contact.
- erpSource is set to MANUAL or the UI source value.
- If VAT/TIN provided, PEPPOL participant ID auto-derived as 0190:ZW{vat}.
- The customer is immediately searchable by company name and erpCustomerId.

B-02: Create customer with multiple contacts via UI

Priority	High	Feature Ref	Flow B
Description	Verify: create customer with multiple contacts via ui	Status	Pass / Fail / Blocked

Test Steps:

37. 1. Navigate to Customers page
38. 2. Click "Register Customer"
39. 3. Fill company details
40. 4. Add Contact 1 (email, name, phone)
41. 5. Click "Add another contact"
42. 6. Add Contact 2 (different email)
43. 7. Save

Expected Result:

- Customer is created with both contacts linked.
- Customer detail panel shows both contacts listed.
- Both contacts are searchable by email.
- Duplicate erpCustomerId is rejected with error message.

B-03: Edit customer details via detail panel

Priority	Medium	Feature Ref	Flow B
Description	Verify: edit customer details via detail panel	Status	Pass / Fail / Blocked

Test Steps:

44. 1. Click a customer row to open detail panel
45. 2. Edit company name, trading name, delivery mode
46. 3. Save changes
47. 4. Close and re-open detail panel

Expected Result:

- Changes are saved and persisted.
- Re-opening the detail panel shows updated values.
- Updated timestamp is refreshed.
- Contacts are unaffected by customer-level changes.

B-04: Add contact to existing customer via detail panel

Priority	Medium	Feature Ref	Flow B
Description	Verify: add contact to existing customer via detail panel	Status	Pass / Fail / Blocked

Test Steps:

48. 1. Open customer detail panel
49. 2. Click "Add Contact"
50. 3. Fill email, name, phone
51. 4. Save

Expected Result:

- New contact is added to the customer.
- Contact appears in the contacts list.
- Duplicate email (same system) is rejected.
- New contact would be CC'd on future invoice sends to this customer.

Flow C: Customer Import via CSV

C-01: Import customers via CSV — basic flow

Priority	Critical	Feature Ref	Flow C
Description	Verify: import customers via csv — basic flow	Status	Pass / Fail / Blocked

Test Steps:

52. 1. Prepare a CSV file with columns: erpCustomerId, companyName, email, name
53. 2. Navigate to Customers page
54. 3. Click "Import CSV"
55. 4. Upload the CSV file (drag-and-drop or file picker)
56. 5. Confirm the auto-suggested column mapping on Step 2
57. 6. Submit

Expected Result:

- CSV import wizard opens with 3 steps: Upload → Map → Results.
- Step 1 shows file name and size.
- Step 2 shows detected columns with auto-suggested mapping; erpCustomerId is required.
- Step 3 shows summary: totalRows, created, updated, skipped, errors.
- Each unique erpCustomerId creates one Customer record.
- Each email creates one Contact linked to the Customer.
- Customers appear in the list after import.

Notes: Download sample CSV template if available to match expected format.

C-02: CSV import with numeric delivery mode mapping (1=EMAIL, 2=AS4, 4=BOTH)

Priority	High	Feature Ref	Flow C
Description	Verify: csv import with numeric delivery mode mapping (1=email, 2=as4, 4=both)	Status	Pass / Fail / Blocked

Test Steps:

58. 1. Prepare CSV with deliveryMode column containing values 1, 2, and 4
59. 2. Import via CSV wizard
60. 3. Check created customers

Expected Result:

- Customers with deliveryMode=1 have EMAIL mode.
- Customers with deliveryMode=2 have AS4 mode.
- Customers with deliveryMode=4 have BOTH mode.
- Invalid/missing values inherit the organisation's default delivery mode.

C-03: CSV import upserts by erpCustomerId (dedup)

Priority	Critical	Feature Ref	Flow C
Description	Verify: csv import upserts by erpcustomerid (dedup)	Status	Pass / Fail / Blocked

Test Steps:

61. 1. Import CSV with erpCustomerId=TEST-UPSERT-001, companyName="Company A"
62. 2. Edit CSV to change companyName to "Company A Updated"
63. 3. Re-import same erpCustomerId=TEST-UPSERT-001

Expected Result:

- Only one Customer record exists for erpCustomerId=TEST-UPSERT-001.
- Company name reflects the latest import ("Company A Updated").
- Contacts are upserted by email: new emails added, existing emails updated.
- Import result shows "updated" count incremented, "created" count unchanged.

C-04: CSV import with multiple contacts per row (email2, name2, phone2)

Priority	Medium	Feature Ref	Flow C
Description	Verify: csv import with multiple contacts per row (email2, name2, phone2)	Status	Pass / Fail / Blocked

Test Steps:

64. 1. Prepare CSV with columns: erpCustomerId, companyName, email, name, email2, name2, phone2, email3, name3
65. 2. Import
66. 3. Check the customer detail panel

Expected Result:

- One Customer record created.
- Three Contact records linked to the customer (email, email2, email3).
- Each contact has the correct name and phone populated.
- All contacts appear in the customer detail panel.

Notes: Supports up to 4 contact tuples per row (email2-4, name2-4, phone2-4).

C-05: CSV import preview before actual import

Priority	Medium	Feature Ref	Flow C
Description	Verify: csv import preview before actual import	Status	Pass / Fail / Blocked

Test Steps:

67. 1. Prepare a CSV with valid data
68. 2. Use the import preview feature (POST /api/v1/organizations/{orgId}/customers/import/preview)
69. 3. Review preview output

Expected Result:

- Preview returns column headers detected.
- Shows up to 5 sample rows with values.
- Shows the suggested column mapping.
- No data is actually created in the database.
- HTTP 200 returned.

C-06: CSV import error handling — bad row does not abort entire import

Priority	Medium	Feature Ref	Flow C
Description	Verify: csv import error handling — bad row does not abort entire import	Status	Pass / Fail / Blocked

Test Steps:

70. 1. Prepare CSV with 10 rows; row 5 has an invalid email and row 7 is missing erpCustomerId
71. 2. Import
72. 3. Check import results

Expected Result:

- Import completes with results for all rows.
- errors count > 0.
- Rows 1-4, 6, 8-10 are successfully imported.
- Row 5 shows error: "Invalid email format".
- Row 7 shows error: "erpCustomerId is required".
- Previously imported rows are NOT rolled back.
- Error messages are descriptive.

C-07: CSV import with custom column mapping (non-standard column names)

Priority	Medium	Feature Ref	Flow C
Description	Verify: csv import with custom column mapping (non-standard column names)	Status	Pass / Fail / Blocked

Test Steps:

73. 1. Prepare CSV with non-standard column names: "CustomerCode", "LegalName", "EmailAddress"
74. 2. Upload via import wizard
75. 3. On Step 2 (Map), manually map:

- CustomerCode → erpCustomerId
- LegalName → companyName
- EmailAddress → email

76. 4. Submit

Expected Result:

- Custom mapping is accepted.
- Each mapping dropdown shows source column values preview.
- erpCustomerId mapping is required (cannot proceed without it).
- Import succeeds with correct field mapping.
- Customer records reflect the mapped fields correctly.